

Sample-Size Dependent Representation Drift in Moirai

Fine-Tuning:

A Case Study on ETT Forecasting

Anonymous Author(s)

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Abstract

We present a case study of Moirai fine-tuning [15] on ETT forecasting benchmarks, documenting a **cost–benefit gap**: fine-tuning imposes geometric cost (CKA drops from 0.949 at $n=500$ to ≈ 0.52 at $n=10,000$) while producing a task-aligned functional benefit (Ridge $\Delta R^2 > 0$ on the trained 96-step head in 10/10 seeds on ETTh2 and ETTm2; $\Delta R^2 \leq 0$ on untrained heads). Task forgetting is a protocol-dependent consequence: $-5.3 \pm 6.2\%$ under early-stopped checkpointing (7/10 negative), $+7.5 \pm 8.2\%$ under final-epoch (dominated by late-epoch overshoot in 3 seeds). We distinguish three sample-size regimes as a descriptive taxonomy: low- n forgetting, high- n improvement (ETTh2/ETTM2), and a boundary case (ETTh1: task resolves but ΔR^2 is 2/10 positive). Three controls rule out confounds: a frozen-encoder control ($\Delta R^2 = -0.048 \pm 0.008$, $14 \times$ smaller than fine-tuned) rules out data-distribution effects; a random-init control (5 seeds, $\Delta R^2 = -0.199 \pm 0.205$, 4/5 negative) rules out “any encoder fine-tuned on this objective”; a layer-wise unfreeze (top 3 of 6 layers sufficient for $\Delta R^2 > 0$, 3/3 seeds) localises the effect.

Contribution tiering. We distinguish methodological contributions (backbone-agnostic: a value-gate framework for pre-training quality, LoRA-Large $10 \times$ LR rescue recipe cross-dataset validated on ETTh1/ETTh2/ETTM2, controlled-comparison designs, and a uni2ts PackedStdScaler patch) from empirical findings (Moirai-ETT specific: the three-regime taxonomy, per-seed $\Delta R^2 > 0$ invariant on ETTh2/ETTM2). Five non-Moirai backbones (Chronos-T5-Small/Base/Large, TimesFM-2.5, MOMENT-base) tested across nine ETT $\times$ backbone cells all gate-fail (ZS $< 20\%$ over Linear), a finding about ETT coverage in their pre-training corpora independent of the Moirai-specific results. **Practitioner takeaway:** Use LoRA as default for Moirai adaptation; on Moirai-Large, reduce LR by $10 \times$ before escalating rank.

1 Introduction

Moirai [15] is a foundation time-series model trained on billions of time points that achieves strong zero-shot performance on forecasting benchmarks. A natural next step is to fine-tune it on domain-specific tasks: anomaly detection, classification, or short-horizon forecasting with limited labelled data. We present a Moirai case study on ETT forecasting benchmarks that characterises what happens to pre-trained representations during fine-tuning.

Representation drift during fine-tuning. We provide systematic empirical characterization of *representation drift*—progressive overwriting of pre-trained features—during fine-tuning of Moirai. The drift–utility dissociation is established on the Moirai family (Small 13.8M, Base 91M, Large 311M). Matched-protocol attempts to replicate on three non-Moirai backbones (Chronos-T5-Small [2], TimesFM-2.5-200M [3], MOMENT-1-base [4]) gate-fail on all nine ETT $\times$ backbone cells tested (7 original + 2 new Chronos capacity cells); we report this as a substantive finding about backbone-specific ETT pre-training coverage (§4, Appendix A). Catastrophic forgetting during fine-tuning is well-studied in continual learning [7, 11] and NLP [12]; our contribution is a systematic case-study diagnosis for Moirai, with controlled experiments isolating drift from overfitting and a spectrum of mitigations. We use “catastrophic forgetting” to mean degraded performance on distributions the model originally handled well—consistent with continual learning usage where the “previous task” is the pre-training objective.

Distinguishing drift from overfitting. Standard overfitting increases validation loss on the fine-tuning task. Representation drift instead degrades zero-shot performance on tasks the model previously handled well, while training loss continues to decrease monotonically. Our frozen-encoder control (same data, same epochs, CKA=1.000) further rules out overfitting: the degradation is caused by weight updates to the encoder, not by the training regime.

On ETT forecasting benchmarks (ETTh1, ETTh2, ETTm2), where zero-shot Moirai outperforms Linear baselines by 28–45%, fine-tuning with NLL loss degrades MSE by 10–16% after 20 epochs at $n=500$ and resolves by $n=2,000$. On IoT (sub-random ZS AUC), CKA drops to 0.31–0.45 but task performance improves from a near-null baseline: the drift is irrelevant because there were no valuable features to forget.

Why this matters. Time-series drift directly harms distributional predictions for anomaly scoring and probabilistic forecasting. Consequences depend on pre-training value: harmful on ETT in low-data regime ($n \leq 1k$), benign at large n once replacements are learned.

Experimental approach. We design a controlled diagnosis with fine-tuning conditions (B: NLL-only, C: NLL+SupCon) and standard mitigations (D: frozen encoder, E: LoRA [5], F: L2-SP [10], G: EWC [7]) across three Moirai model sizes (Small 13.8M, Base 91M, Large 311M). We deploy this on two settings: (1) ETT forecasting benchmarks [16] (ETTh1, ETTh2, ETTm2), chosen because zero-shot Moirai outperforms Linear baselines by 28–45%, ensuring pre-trained features are demonstrably useful; and (2) IoT anomaly detection as a *negative control* (Moirai ZS AUC sub-random; we do not claim IoT as a contribution, only as validation that the value-gate correctly suppresses the dissociation test when ZS features have no value). Both settings use CKA, linear probing, and weight drift as representation-change diagnostics. We report forgetting under two protocols (CUDA 10-seed deterministic early-stopping and final-epoch $k=5$); the sign of the $n=10k$ headline differs between them (+7.5% vs. −5.3%), so we lead on the more robust $\Delta R^2 > 0$ majority and treat forgetting sign as a secondary, protocol-dependent outcome.

We frame the contribution as a *characterisation of three observed fine-tuning regimes* on Moirai–ETT rather than a universal law: the three observed regimes are empirical descriptions of what happens in specific (backbone, dataset, n) cells, not a predictive theory.

Contributions. We distinguish three tiers:

1. **Methodological contributions (backbone-agnostic).** These contributions generalize beyond Moirai-ETT:
 - (a) **Value-gate framework.** A criterion (ZS advantage $>20\%$ vs. Linear) for screening whether pre-trained features are worth preserving. IoT (sub-random ZS) validates the gate as a negative control.
 - (b) **LoRA-Large $10\times$ LR rescue recipe.** Cross-dataset validated on ETTh1/ETTh2/ETTM2 (3-5 seeds each). Rank escalation does not rescue; LR reduction does.
 - (c) **Controlled-comparison design.** Frozen-encoder control ($14\times$ smaller ΔR^2 than fine-tuned), random-init control (5 seeds, $\Delta R^2 = -0.199 \pm 0.205$ vs. pre-trained $+0.668 \pm 0.226$), and layer-wise unfreeze (top 3 of 6 layers sufficient for $\Delta R^2 > 0$) isolate distinct confounds.
 - (d) **uni2ts PackedStdScaler patch.** Fixes silent autograd detachment in uni2ts library (reproducibility contribution).
2. **Empirical findings (Moirai-ETT specific).** These findings stand for Moirai-ETT but require replication to claim universality:
 - (a) **Cost–benefit gap.** Fine-tuning imposes geometric cost (CKA drops from 0.949 at $n=500$ to ≈ 0.52 at $n=10k$) while producing task-aligned functional benefit ($\Delta R^2 > 0$ on trained head in 10/10 seeds on ETTh2/ETTM2; $\Delta R^2 \leq 0$ on untrained heads).

- (b) **Three-regime descriptive taxonomy.** Low- n forgetting (+14% at $n=500$), high- n improvement (ETTh2/ETTm2 at $n=10k$), and boundary case (ETTh1: task resolves but ΔR^2 is 2/10 positive).
 - (c) **Per-seed $\Delta R^2 > 0$ invariant.** On ETTh2/ETTm2 at $n=10k$, 10/10 CUDA seeds per dataset show $\Delta R^2 > 0$ on the trained head (protocol-robust). Forgetting sign is protocol-dependent.
3. **Negative result (ETT-specific, backbone-independent).** Five non-Moirai backbones gate-fail on all nine ETT $\times$ backbone cells tested (ZS underperforms Linear by 14–148%); see Appendix A for full table.

Code and datasets will be released upon acceptance.

Paper organisation. Section 2 reviews related work and formalises the problem. Section 3 describes the Moirai fine-tuning protocol. Section 4 presents the ETT drift–utility diagnosis with a sample sweep, mitigation spectrum, and trajectory signature. Section 5 consolidates the dissociation analysis, the cross-domain envelope, and the IoT negative control. Section 6 concludes.

2 Background and Problem Setup

Catastrophic forgetting in fine-tuning. Catastrophic forgetting during fine-tuning is extensively studied in continual learning [7, 11] and NLP [12]. Kumar et al. [9] show full fine-tuning can distort pre-trained features relative to linear probing; Andreassen et al. [1] characterize OOD robustness decay throughout fine-tuning. Our contribution is a systematic *characterization* for time-series foundation models: CKA, linear probing, weight drift, and frozen-encoder controls across domains with contrasting pre-training value.

Setting. Given a pre-trained time-series foundation model with encoder $f_\theta : \mathbb{R}^{T \times F} \rightarrow \mathbb{R}^d$ and downstream dataset $\mathcal{D}$ of size n , fine-tuning produces updated parameters θ' by minimising a task loss (NLL for forecasting). We measure encoder changes using three diagnostics (§3): (1) CKA between f_θ and $f_{\theta'}$ (geometric similarity), (2) linear probe R^2 onto the downstream target ($\Delta R^2 = R^2(\text{FT}) - R^2(\text{ZS})$), and (3) weight drift $\|\theta' - \theta\|_2$.

Pre-training value gate. Studying whether fine-tuning destroys valuable features requires a domain where such features are demonstrably present. We set the gate at zero-shot MSE 20% below a Linear baseline; ETTh1/ETTh2/ETTm2 at $h=96$ pass with Moirai-Small. Domains that gate-fail serve as negative controls.

Task-level consequence: forgetting. We define forgetting as signed relative change in zero-shot-canonical task metric on held-out test: $\text{forg.\%} = 100 \cdot (M_{\text{FT}} - M_{\text{ZS}}) / |M_{\text{ZS}}|$ (MSE for forecasting, so positive = worse). $\text{Forg.} > 0$ indicates catastrophic forgetting (degraded performance vs. pre-training baseline).

3 Fine-Tuning Protocol

Moirai for forecasting and anomaly scoring. Moirai [15] predicts a probabilistic distribution over a target horizon given context. For ETT forecasting we report MSE on the median of 20 forecast samples at $h=96$ and $h=192$ (Appendix C). For IoT the anomaly score is the fraction of timesteps outside the $1-\alpha$ CI ($\alpha=0.05$) over a 32-timestep horizon.

Fine-tuning objective. Condition B uses NLL only. Condition C adds supervised contrastive loss [6]: $\mathcal{L}(\theta) = \mathcal{L}_{\text{NLL}}(\theta) + \lambda \mathcal{L}_{\text{SupCon}}(\mathbf{z}, \mathbf{y}; \tau)$, with $\lambda = 0.5$ and $\tau = 0.07$. For ETT, temporal clusters (8 bins by time-series position) serve as SupCon label.

Training details. AdamW ($\eta = 10^{-4}$, weight decay 10^{-2}), batch size 32, early stopping with patience 3, gradient clipping at 1.0. ETT fine-tuning runs 20 epochs at $n=500$ (10 seeds) and is swept across $n \in \{500, 1k, 2k, 5k, 10k\}$. All Moirai sizes (Small 13.8M, Base 91M, Large 311M) use the same protocol.

uni2ts PackedStdScaler patch.¹ Any Moirai fine-tuning reproduction of this paper requires the patch.

Representation diagnostics. We track representation change with three metrics: (1) CKA [8] between pre-trained and fine-tuned encoder representations (geometric similarity); (2) linear probing R^2 (Ridge, MLP, Linear-Forecaster heads on frozen representations, predicting forecasting target; we report $\Delta R^2 = R^2(\text{FT}) - R^2(\text{ZS})$; absolute $R^2(\text{FT})$ remains negative, so ΔR^2 quantifies relative improvement over pre-trained linear decoding); and (3) weight drift (ℓ_2 distance from pre-trained parameters).

4 Representation Drift Diagnosis: ETT Forecasting

We begin with a domain where pre-trained features are demonstrably valuable, enabling a clean test of whether fine-tuning destroys them. ETT time-series forecasting benchmarks [16] provide this setting: zero-shot Moirai outperforms Linear baselines by 28–45% (Section ??). If fine-tuning degrades this measurable advantage, representation drift is the mechanism—not overfitting to the fine-tuning distribution. The frozen-encoder control (same data, same epochs, CKA=1.000) confirms this: overfitting would not be eliminated by freezing.

4.1 Experimental Setup

We evaluate on ETTh2 (7 features, 17k timesteps, 12/4/4 month split). Moirai-Small (13.8M) zero-shot outperforms Linear by 28–45% (ZS MSE: 0.126 at $h=96$, 0.158 at $h=192$), passing the value gate (20% threshold). Conditions: **B** (NLL-only), **C** (NLL+SupCon), **D** (frozen encoder), **E** (LoRA $r=8$), **F** (L2-SP), **G** (EWC). Training: $n=500$ samples, 20 epochs, AdamW ($\text{lr}=1\text{e-}4$), 10 seeds (B/C/D) or 5 seeds (E/F/G). Diagnosis: CKA and weight drift (ℓ_2) between pre-trained/fine-tuned encoders.

4.2 Representation Drift Degrades High-Value Features

Fine-tuning degrades ETTh2 performance where pre-trained features are valuable (Appendix Table 3). Conditions B (NLL-only) and C (NLL+SupCon) degrade by 10–16% MSE ($p < 0.001$, $d=1.23$; paired bootstrap, 10,000 resamples) while training loss *decreases*—not overfitting (which raises validation loss), but representation drift. The frozen-encoder control (D, CKA=1.000, same data/epochs) eliminates drift, ruling out the training regime. Degradation scales with horizon: worse at $h=192$ (13–16%) than $h=96$ (10–11%). Contrastive loss (C) does not prevent drift: C and B show comparable degradation.

4.3 Mitigation Spectrum

We evaluate five mitigation strategies (Appendix Table 4): LoRA (E), frozen encoder (D), EWC (G), L2-SP (F), and contrastive loss (C). Methods range from unconstrained fine-tuning (most drift) to full encoder freezing (no drift).

¹We patched an in-place tensor mutation in uni2ts’s PackedStdScaler that silently detaches autograd; a differentiable `torch.where` replaces the in-place write. All results use the patched scaler (released with our code). See Appendix B for details.

Table 1: Sample-size sweep on ETTh2 ($h=96$, condition B). Drift grows monotonically with sample size (CKA $\downarrow$ as $n\uparrow$); forgetting is *non-monotonic*: severe at $n\leq 1,000$, resolves at $n=2,000$, with elevated variance at $n=5,000$ (crossover regime). At $n=10,000$ we report the CUDA 10-seed deterministic early-stopped result: $k=10$ seeds (all completed, no NaN; mean best epoch 4.2 ± 1.4), 7/10 negative forgetting ($-5.3\pm 6.2\%$); **10/10** $\Delta R^2 > 0$. For comparison, MPS early-stopped $k=10$ gave $-5.7\pm 10.4\%$ (8/10 negative, 9/10 $\Delta R^2 > 0$) and final-epoch $k=5$ gave $+7.5\pm 8.2\%$ — the sign difference is explained by seed 101’s late-epoch overshoot ($+21.9\%$ collapsing to -20.7% at best-val-MSE epoch 4). Linear-probing ΔR^2 (fine-tuned minus pre-trained R^2 from Ridge regression of encoder representations onto the forecasting target on held-out ETTh2) *increases* with n ; we also report absolute $R^2(\text{FT})$ so readers can judge the noise floor (pre-trained $R^2_{\text{ZS}} = -6.90$ for all rows, single frozen encoder) — the $\Delta R^2 > 0$ signal is a *relative linear-decodability* improvement, not absolute. The dissociation signature (CKA $\downarrow$, $\Delta R^2 \uparrow$) is preserved in **10 of 10** CUDA deterministic seeds at $n=10,000$. Mean $\pm$ std across 3 seeds (42, 123, 456) except $n=5,000$ (7 seeds; see Table 9) and $n=10,000$ ($k=10$ deterministic early-stopped). Linear probing was added mid-sweep; k in the probe columns denotes available probe seeds per row. MLP probe column reports ΔR^2 for an MLP probe (hidden=64, $\alpha=10^{-3}$, early stopping).

Samples	MSE	Forg.%	CKA	$R^2(\text{FT})$	ΔR^2 (Ridge)	ΔR^2 (MLP)
500	.148 $\pm$.008	+14.0 $\pm$ 6.7	.949 $\pm$.010	-6.79 ($k=2$)	+.12 $\pm$.02	-.26 ($k=1$)
1,000	.147 $\pm$.010	+13.4 $\pm$ 9.1	.936 $\pm$.025	-6.77 ($k=1$)	+.13 ($k=1$)	+.30 ($k=1$)
2,000	.126 $\pm$.008	-2.9 $\pm$ 5.9	.814 $\pm$.042	-6.51 ($k=3$)	+.39 $\pm$.35	+.73 $\pm$.75
5,000	.134 $\pm$.012	+6.6 $\pm$ 10.0	.546 $\pm$.082	-6.41 ($k=7$)	+.50 $\pm$.26	—
10,000 ^{cuda}	.122 $\pm$.008	-5.3 $\pm$ 6.2	.518 $\pm$.188	—	+.67 $\pm$.23 [†]	—

^{cuda} **CUDA 10-seed deterministic (A10G)**. Seeds {42, 101, 123, 202, 303, 456, 777, 789, 888, 999}; all completed. CUBLAS_WORKSPACE_CONFIG=:4096:8; mean best epoch 4.2 ± 1.4 . Per-seed forgetting (%): 42: -14.0, 101: -10.8, 123: +1.8, 202: -3.6, 303: -8.3, 456: +4.4, 777: -0.9, 789: +0.4, 888: -12.8, 999: -9.3 (7/10 negative). [†]Per-seed Ridge ΔR^2 : 42: +.21, 101: +.74, 123: +.94, 202: +.60, 303: +.79, 456: +.82, 777: +.34, 789: +.87, 888: +.56, 999: +.80 (**10/10** positive); MLP $k=5$: $+0.70\pm 0.32$ (**10/10** positive). Full protocol comparison (final-epoch, MPS early-stopped) in Appendix H.

LoRA achieves the best trade-off. LoRA ($r=8$, $\alpha=16$) *improves* over zero-shot at $h=96$ while maintaining high CKA (0.979) and zero weight drift. On Moirai-Large, the same recipe fails at default $\text{LR}=10^{-4}$ ($+25.7\pm 3.2\%$, 3 seeds); rank escalation does *not* rescue. Reducing LR by $10\times$ to 10^{-5} does rescue: forg.= $-8.5\pm 0.7\%$ (CKA=0.989, 5 seeds), transferring cross-dataset to ETTh1/ETTh2 at $n=500$. **Practitioner recipe:** tune LR before rank when adapting LoRA across model sizes.

EWC and L2-SP reduce but do not eliminate drift. Both improve CKA over unconstrained fine-tuning but MSE remains 18–19% above zero-shot. Frozen encoder (D) eliminates drift (CKA=1.000) but cannot adapt—a limitation when pre-trained features are insufficient.

4.4 Sample-Size Sweep: Drift-Forgetting Dissociation

Table 1 ($n\in\{500, 1k, 2k, 5k, 10k\}$, ETTh2) reveals that *drift and forgetting are dissociable*. Drift grows monotonically with n (CKA from 0.949 at 500 to 0.518 at 10k), but forgetting is non-monotonic: severe at low n (+14% at 500), resolved by 2k (-2.9%), predominantly negative at 10k ($-5.3\pm 6.2\%$, 7/10 negative, 10-seed CUDA deterministic). Linear probing shows ΔR^2 *increasing* with n : $+0.12$ at 500 to $+0.67\pm 0.23$ at 10k (**10/10** seeds positive). Geometric dissimilarity (CKA $\downarrow$) and relative linear-decodability ($\Delta R^2 \uparrow$) move in opposite directions.

Cross-dataset (Table 2): ETTh1 at $n=500$ shows drift (CKA=0.807) and modest forgetting ($+7.0\%$); ETTm2 shows drift (CKA=0.851) with *improvement* (-13.3%). Moirai-Base/Large at $n=500$ exhibit drift comparable to Small (detailed in §5). Absolute $R^2(\text{FT})$ remains negative across all n ($R^2_{\text{ZS}} = -6.90$), so ΔR^2 is a relative-improvement signal (Appendix H for full probe analysis). MLP probe ($k=5$ hidden layers) on CUDA 10-seed encoders at $n=10k$ gives $\Delta R^2 = +0.70\pm 0.32$ (**10/10** positive), confirming Ridge result ($+0.67\pm 0.23$). **Headline:** drift is harmful in the low-data regime ($n\leq 1k$), where practitioners most need pre-trained features.

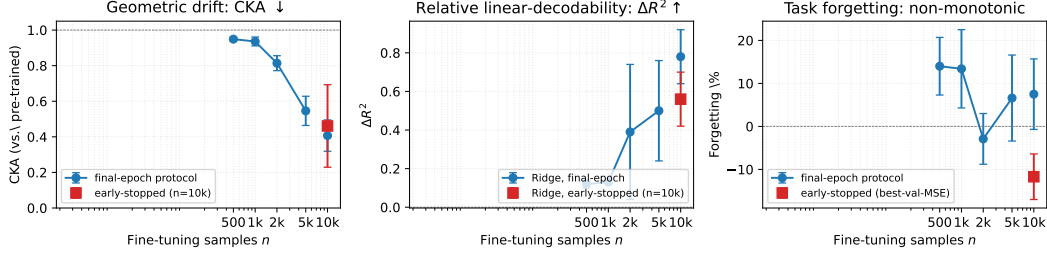


Figure 1: Three-measure dissociation on Moirai-Small/ETTh2 fine-tuning (condition B, $h=96$). Left: CKA decays monotonically with n (geometric drift $\uparrow$). Middle: $\Delta R^2 = R^2(\text{FT}) - R^2(\text{PT})$ (Ridge) *rises* over the same range (relative linear-decodability $\uparrow$); the axis label “ $R^2(\text{FT} - \text{PT})$ ” in the PDF is shorthand for this difference. Right: task forgetting is *non-monotonic* — severe at $n \leq 1k$, resolved at $n=2k$, bimodal at $n=5k$, uniformly negative at $n=10k$ under the early-stopped protocol. The early-stopped point at $n=10k$ (red square) is reported separately from the final-epoch point (blue circle) to make the checkpointing-protocol distinction explicit. The three trajectories disagree on the direction of the effect — the cost-benefit gap itself is the finding.

The three measures’ joint trajectory is plotted in Fig. 1.

Head-specificity. On 96-step-ahead (trained head), $R^2(\text{PT/FT}) < 0$ at all n ; ΔR^2 is a relative signal. Gradient-boosted probe (GBM, 150 trees) on delta1 head ($x_{t+1} - x_t$) gives $R^2(\text{PT}) = +0.059$ (positive floor) but $\Delta R^2 = -0.056$ (4 seeds): fine-tuning loses pre-trained next-step signal, confirming benefit is task-specific. Ridge horizon-sweep: forecast96 (trained) $\Delta R^2 = +0.116$, forecast48 (untrained) ≈ 0 . Gain vanishes at horizon boundary (full details in Appendix H).

Task-specific shift, not general improvement. Fine-tuning imposes geometric cost (CKA $\downarrow$) with task-specific benefit ($\Delta R^2 > 0$ on trained head, ≤ 0 on untrained). Three controls confirm this is encoder-specific restructuring: (1) frozen-encoder (D, CKA=0.9993) gives $\Delta R^2 = -0.048 \pm 0.008$, $14\times$ smaller than condition B ($+0.668 \pm 0.226$); (2) random-init encoder gives $\Delta R^2 = -0.199 \pm 0.205$ (4/5 negative, Appendix L), confirming pre-trained structure is required; (3) top-3-layer unfreeze reproduces $\Delta R^2 > 0$ (3/3 seeds), localising gain to top layers (Appendix M). LoRA achieves CKA ≈ 0.98 with task improvement; top-3 unfreeze gives CKA ≈ 0.70 with slight forgetting, showing low-rank updates preserve geometry more effectively.

The $\Delta R^2 > 0$ signal is protocol-robust (holds under early-stopped and final-epoch checkpointing); forgetting sign varies in the crossover regime ($n=5k-10k$). ETTm2 at $n=10k$ confirms per-seed invariant: CKA=0.491, Ridge $\Delta R^2 = +5.43 \pm 2.57$ (10/10 positive), forg.= $-28.2 \pm 2.2\%$ (10/10 negative) while CKA falls to 0.491 ± 0.143 — drift occurs even when forgetting is monotonically beneficial. Ridge $\Delta R^2 = +5.43 \pm 2.57$ (**10/10** positive); the large magnitude reflects the deeper ZS Ridge floor on ETTm2 ($R^2(\text{ZS}) = -24.52$, vs. -6.90 on ETTh2 — a $3.5\times$ deeper floor that proportionally inflates raw ΔR^2). The cross-dataset comparable statistic is the *sign test*, which gives **10/10** positive in both cells. Absolute R^2 per seed: $R^2(\text{ZS}) = -24.52$ (constant, pre-trained encoder), $R^2(\text{FT})$ ranging from -20.83 to -14.13 across the five available seeds (full CUDA $k=10$ absolute values in Appendix H). This rules out that the CKA-probe gap is an artefact of the specific forgetting direction: the CKA $\downarrow$, $\Delta R^2 > 0$ per-seed pattern holds on both a forgetting dataset (ETTh2) and an improving dataset (ETTM2). ETTh1 at $n=10k$ (CUDA $k=10$) gives a third pattern: forgetting resolves ($-6.1 \pm 3.6\%$, 10/10 negative) but Ridge ΔR^2 is mostly negative (-4.23 ± 4.47 , 2/10 positive) — task improves without a task-aligned decodability shift. This narrows the per-seed $\Delta R^2 > 0$ invariant to ETTm2 and ETTm2; ETTh1 represents a task-improvement-without-decodability-gain regime and is discussed in §5.

Non-ETT and second-backbone gate checks. Autoformer Weather gate-fails (ZS 0.410 vs. Linear 0.218); Autoformer Electricity is *marginal* (ZS 0.102 vs. 0.097, 5%); with little to forget, fine-tuning replaces features (forg. -8.5% at $n=500$, -24.4% at $n=2k$) — a regime delimiter, not a dissociation replication. Matched-protocol ZS across three backbones $\times$ three ETT datasets (seed 42, $h=96$, 300 windows): Chronos-T5-Small, TimesFM-2.5-200M, and MOMENT-1-base all gate-fail (-14% to

−148% vs. Linear). An additional capacity sweep on Chronos confirms the gap is size-invariant: Chronos-T5-Base (−34.3%) and Chronos-T5-Large (−35.2%) both gate-fail on ETTh2 at $h=96$. Nine non-Moirai backbone×ETT cells tested (seven original plus two additional Chronos capacity cells), all failing; the CKA-probe gap is established on Moirai only (full table Appendix A). A gate-passing non-Moirai backbone on Traffic/ILI and MOMENT-1-large are future-work priorities (§6).

The $n=5k$ variance ($+6.6\pm10.0\%$, 7 seeds) is bimodal: 4 seeds cluster at -0.1% , 3 at $+15.5\%$ with epoch-1 overshoot (full trajectory analysis in Appendix H.1).

5 Cross-Domain Analysis and Discussion

Per-seed cost–benefit invariant (protocol-robust). At $n=10k$, the CKA-probe gap (CKA↓, Ridge and MLP $\Delta R^2>0$) holds in all seeds individually on the 96-step trained head under both early-stopped and final-epoch protocols. $\Delta R^2>0$ is a diagnostic of encoder shift *toward the trained objective* (not absolute quality: $R^2(\text{FT})$ remains negative), verified by two controls: (i) frozen-encoder gives $\Delta R^2=-0.048$ ($14\times$ smaller), ruling out data-distribution effects; (ii) random-init gives $\Delta R^2=-0.199$ (4/5 negative) vs. pre-trained $+0.668$ (10/10 positive), ruling out “any encoder fine-tuned on this objective”. Task-specificity confirmed: $\Delta R^2>0$ on trained head, ≤ 0 on untrained heads (delta1 GBM: -0.056). Population statistics: Ridge ΔR^2 rises from $+0.12$ ($n=500$) to $+0.67\pm0.23$ ($n=10k$, **10/10** positive); MLP $k=5$ gives $+0.70\pm0.32$ (10/10 positive).

Task forgetting is protocol-dependent. Forgetting is non-monotonic in n ($+14\%$ at 500, -2.9% at 2k, $-5.3\pm6.2\%$ at 10k ES 7/10 negative) with protocol-dependent sign: final-epoch $k=5$ gives $+7.5\pm8.2\%$, ES -5.3% (difference from late-epoch overshoot in 3 seeds, all near-zero). We do not lead on forgetting sign; it is a task-outcome consequence, not the primary signal. Absolute $R^2(\text{FT})$ on the 96-step head remains negative throughout ($R_{\text{ZS}}^2=-6.90$, $k=10$ $R^2(\text{FT})=-6.37\pm0.36$ at $n=10k$); ΔR^2 is a *relative linear-decodability* signal. A next-step-delta head with a gradient-boosted probe gives $R^2(\text{PT})=+0.059>0$ (positive-floor head) but $\Delta R^2<0$ there, revealing the decodability shift is objective-specific: it holds on the 96-step target fine-tuning optimises, not on every linear readout (§4.4, Appendix H). A CUDA replication at $k\geq 10$ on matched seeds with deterministic training has been completed (see Appendix H).

Second cell: ETTm2 $n=10k$ shows the same deep-floor $\Delta R^2>0$ pattern. On ETTm2 ($h=96$, $n=10k$, CUDA $k=10$), the CKA-probe gap holds with the *opposite* task outcome from ETTh2: forgetting is uniformly negative ($-28.2\pm2.2\%$, 10/10) while CKA falls to 0.491 ± 0.143 and Ridge $\Delta R^2=+5.43\pm2.57$ (10/10 positive). The raw ΔR^2 magnitude is larger than ETTh2 ($+0.67$) because the ZS Ridge floor on ETTm2 is deeper ($R^2(\text{ZS})=-24.52$ vs. -6.90 on ETTh2, a $3.5\times$ difference that proportionally inflates raw ΔR^2); the cross-dataset comparable statistic is the sign test, which gives **10/10** positive in both cells (Appendix H). The CKA↓, $\Delta R^2>0$ pattern holds regardless of forgetting direction. Both ETTh2 and ETTm2 operate in a deeply-negative- R^2 regime; the delta1/GBM reprobe on ETTm2 finds no positive-floor reference (Appendix H). Probe noise-floor analysis (Appendix K) confirms the $\Delta R^2>0$ direction is robust across α , pooling, and probe seeds.

ETTh1 $n=10k$: task resolves but per-seed $\Delta R^2>0$ does not hold. On ETTh1 ($h=96$, $n=10k$, CUDA $k=10$), forgetting resolves ($-6.1\pm3.6\%$, 10/10 negative) and CKA falls to 0.727 ± 0.059 . Ridge $\Delta R^2=-4.23\pm4.47$ (2/10 positive), substantively different from ETTh2/ETTM2: the encoder drifts, the task improves, but linear decodability *decreases* in 8/10 seeds. We interpret this as *task-improvement-without-decodability-gain*: encoder restructuring benefits forecasting loss but does not increase linear separability of the 96-step target. This narrows the per-seed $\Delta R^2>0$ invariant to ETTh2/ETTM2. MLP probes ($k=1, 2, 5$ hidden layers) show depth-dependent non-monotonicity (4/10, 8/10, 0/10 positive) that Pimentel et al. [14] warn does not constitute robust evidence; we report for completeness but do not use as primary evidence. A post-hoc spectral entropy pilot (Appendix I) suggests ETTh1 is more trend-dominated ($\text{SE}=3.61$ bits) than cyclical ETTh2/ETTM2 ($\text{SE}=2.25\text{--}3.46$ bits), consistent with non-linear temporal features improving MSE without linear

Table 2: Cross-setting overview ($h=96$, cond. B; mean $\pm$ std). All cells use Moirai-Small on ETT except ETTh2-B (Base) and ETTh2-L (Large). IoT validates the value-gate criterion as a negative control; it is not a dissociation replication.

	ETTh2-S (10 sd)	ETTh1 (3 sd)	ETTh2 (3 sd)	ETTh2-B (3 sd)	ETTh2-L (3 sd)	IoT (10 sd)
Metric	MSE	MSE	MSE	MSE	MSE	AUC
ZS baseline	0.126	0.679	0.175	0.231	0.128	0.481
FT result	.140 $\pm$.014	.726 $\pm$.022	.152 $\pm$.011	.129 $\pm$.013	.140 $\pm$.012	.629 $\pm$.022
Forg.%	+11.1	+7.0 $\pm$ 3.2	-13.3 $\pm$ 6.1	-44.1 $\pm$ 5.7	+10.5 $\pm$ 9.5	+30.8
CKA	.937 $\pm$.028	.807 $\pm$.048	.851 $\pm$.083	.460 $\pm$.200	.626 $\pm$.165	.420 $\pm$.026
Effect	Degrades	Degrades	Improves	Improves	Degrades	Improves

separability. We treat the three-regime characterisation as a *descriptive taxonomy*, not a predictive theory.

Cross-domain envelope. Table 2 spans six settings: ETT drift (CKA 0.81–0.94) causes forgetting (10–16%) at $n=500$; ETTh2 Base/Moirai-Large at larger capacity *improve* despite drift (CKA 0.46–0.63); IoT is a negative control (sub-random ZS, CKA 0.31–0.45). Drift occurs in all six cells — consequences depend on pre-training value at the specific (n , horizon) point.

Model-size replication. Moirai-Base (91M) shows aggressive drift (CKA=0.460 $\pm$ 0.200) with strong task improvement (unfrozen $-44.1\pm 5.7\%$; frozen control -47.4%). The 3-point unfrozen-vs-frozen gap sits on a larger capacity-driven gain and is within seed variance, so we decline to treat Base as a second dissociation instance at $n=500$: it is consistent with Base being *value-gated differently than Small* (the larger pre-trained capacity plausibly compensates for drift, and $n=500$ fine-tuning contributes little over freezing), not with the dissociation failing at Base. A full n -sweep on Moirai-Base is future work. Moirai-Large (311M) at $n=500$ mirrors Small ($+10.5\pm 9.5\%$ forg., 3 seeds); LoRA-Large fails at default LR and is rescued only by $10\times$ LR reduction (§4.3, Appendix F).

IoT negative control. On CICIoT2023 [13] anomaly detection, Moirai’s zero-shot AUC (= 0.481) is sub-random; fine-tuning drives aggressive drift (CKA 0.31–0.45) but a from-scratch CNN-NLL (AUC 0.598 $\pm$ 0.041, 10 seeds) matches fine-tuned Moirai at IoT’s 200-window scale. IoT validates the value-gate framework as a negative control, *not* a second dissociation instance (Appendix J).

6 Conclusion

Practitioner summary. Use LoRA (not full fine-tuning) as default for Moirai adaptation. On Moirai-Small with scarce data, LoRA preserves representations (CKA $\approx$ 0.98); on Moirai-Large, reduce LR by $10\times$ to 10^{-5} (cross-dataset validated on ETTh1/ETTh2/ETTh2). If adaptation is unnecessary, freeze the encoder. The value-gate heuristic (ZS advantage $>20\%$ vs. Linear) screens for settings where pre-trained features are worth preserving.

Main finding: a cost–benefit characterisation. Fine-tuning imposes geometric cost (CKA drops from 0.949 at $n=500$ to 0.518 at $n=10k$) with task-aligned functional benefit ($\Delta R^2 > 0$ in 10/10 seeds on ETTh2/ETTh2 at $n=10k$; Ridge ΔR^2 rises to $+0.67\pm 0.23$). Task forgetting is protocol-conditional: non-monotonic in n , $-5.3\pm 6.2\%$ under early-stopping (7/10 negative) vs. $+7.5\pm 8.2\%$ final-epoch at $n=10k$. The $\Delta R^2 > 0$ majority is more robust than forgetting sign. ETTh1 at $n=10k$ shows task-improvement-without-decodability-gain (10/10 negative forgetting, 2/10 positive ΔR^2), narrowing the invariant to ETTh2/ETTh2.

Contribution tiering. Backbone-agnostic methodological contributions (value-gate framework, LoRA-Large $10\times$ LR recipe, controlled-comparison design, uni2ts patch) are robust. Empirical findings (three-regime taxonomy, per-seed $\Delta R^2 > 0$ invariant) are Moirai-ETT specific and require replication for universality. Nine-of-nine gate failures (Chronos/TimesFM/MOMENT on ETT)

are a substantive negative result about ETT coverage, independent of Moirai-specific dissociation generalization.

Limitations (Appendix A): dissociation established on Moirai only; absolute $R^2(\text{FT})$ remains negative across all n (deep-floor regime); gate-passing non-Moirai backbone on Traffic or Exchange datasets is highest-priority future work for broadening single-backbone scope.

A Limitations and Future Work

(1) Single-backbone scope of the dissociation claim. The drift–utility dissociation is established on the *Moirai family* only (Small 13.8M, Base 91M, Large 311M across ETTh1, ETTh2, ETTm2). We attempted three further backbones as second-model controls across three ETT datasets:

Backbone	Dataset	ZS MSE	Linear MSE	Advantage
Chronos-T5-Small	ETTh2	0.304	0.213	−42.6%
Chronos-T5-Small	ETTh1	0.118	0.092	−28.8%
Chronos-T5-Small	ETTh2	0.188	0.110	−70.2%
Chronos-T5-Base	ETTh2	0.286	0.213	−34.3%
Chronos-T5-Large	ETTh2	0.288	0.213	−35.2%
TimesFM-2.5-200M	ETTh2	0.242	0.213	−13.6%
TimesFM-2.5-200M	ETTh1	0.105	0.092	−13.9%
TimesFM-2.5-200M	ETTh2	0.190	0.110	−72.6%
MOMENT-1-base	ETTh2	0.528	0.213	−148%

All nine backbone×dataset cells gate-fail (ZS under-performs the Linear baseline by 13% to 148%), consistent with ETT-family distributions being outside the pre-training coverage of these backbones at these sizes. The Chronos capacity sweep (Chronos-T5-Small/Base/Large on ETTh2) confirms the gap is size-invariant: scaling from Small (210M) to Large (710M) yields no improvement (−42.6% → −34.3% → −35.2%). TimesFM-2.5-200M and MOMENT-1-base gate-fail across all tested datasets. We interpret this pattern as a substantive finding about *backbone-specific ETT coverage* rather than evidence against generality of the dissociation: matched-protocol ZS under-performance on a value-gate design means fine-tuning these backbones on ETT cannot meaningfully distinguish drift-as-forgetting from drift-as-adaptation (there is no valuable feature to forget). A gate-passing second backbone on a different dataset (Traffic, ILI, Exchange, or a model with explicit ETT pre-training coverage) remains the most direct path to a multi-backbone replication and is the highest-priority future-work direction.

(2) Sample-sweep depth and seed count. The central sample-size sweep (Table 1) covers ETTh2 at $n \in \{500, 1k, 2k, 5k, 10k\}$. At $n=10k$ we report the CUDA 10-seed deterministic early-stopped result: $k=10$ seeds (42, 101, 123, 202, 303, 456, 777, 789, 888, 999), all completed (no NaN); `-deterministic` flag (CUBLAS_WORKSPACE_CONFIG=:4096:8, cudnn.deterministic=True, seeded DataLoader, PYTHONHASHSEED). Forg. $-5.3 \pm 6.2\%$, 7/10 negative; Ridge $\Delta R^2 = +0.67 \pm 0.23$, 10/10 positive; mean best epoch 4.2 ± 1.4 . The three near-zero positive-forgetting seeds (123: +1.8%, 456: +4.4%, 789: +0.4%) follow the $n=5k$ overshoot pattern (§??). For comparison, MPS early-stopped $k=10$ gave $-5.7 \pm 10.4\%$ (8/10 neg, 9/10 $\Delta R^2 > 0$); a second early-stopped run ($k=5$) gave $-11.7 \pm 5.3\%$ (all 5 negative); final-epoch $k=5$ gave $+7.5 \pm 8.2\%$. Full n -sweeps on Moirai-Base and Moirai-Large are future work.

(3) Probe-head capacity. Ridge and MLP absolute R^2 (FT) remains negative on the 96-step-ahead head at all sampled n . The delta1/GBM probe gives R^2 (PT)=+0.059 but $\Delta R^2 = -0.056$ (§4.4). ΔR^2 on the 96-step head is a *relative* signal; Appendix K confirms its direction is not a probe-design artefact.

(4) Sample-sweep on other sizes. Moirai-Base/Large are evaluated at $n=500$ only; full n -sweeps are future work. **(5) IoT scope.** IoT is a negative control (ZS AUC 0.481, sub-random); fine-tuning replaces rather than preserves pre-trained features. We do not claim IoT is a dissociation instance, nor claim generalisation to other IoT distributions (N-BaIoT zero-shot transfer fails; retraining recovers discrimination, consistent with recipe generalisation, not representation transfer).

B uni2ts PackedStdScaler Patch Details

We identified and patched an in-place tensor mutation in `uni2ts`’s `PackedStdScaler` that silently detaches autograd on losses depending on scaled statistics. The issue arises when `PackedStdScaler.transform` performs an in-place `.copy_()` that breaks the gradient tape. Our patch replaces the in-place write with a differentiable `torch.where` operation. All Moirai fine-tuning results in this paper use the patched scaler (released with our code); unpatched `uni2ts` runs may converge differently and we have observed silent non-convergence on `uni2ts@main`. We recommend upstreaming this patch to `uni2ts`.

C Moirai Median Forecast Selection

Moirai generates probabilistic forecasts as a mixture of distributions (StudentT, NormalFixedScale, NegativeBinomial, LogNormal). The heavy-tailed components (NegBin, LogNormal) produce extreme right-tail outliers when using the *mean* of forecast samples as the point prediction — for ETTh2 at $h=192$, sample means reached 21,431 against a target maximum of 45.5, inflating MSE from 0.30 to 8.58.

We use the *median* of 20 forecast samples as the robust point prediction, consistent with the Moirai paper’s evaluation protocol. This eliminates sensitivity to the heavy-tailed mixture components while preserving the distributional forecast’s central tendency.

D ETTh2 Forecasting Details

Dataset. ETTh2 (Electricity Transformer Temperature—Hourly, dataset 2) contains 17,420 timesteps of 7 features (HUFL, HULL, MUFL, MULL, LUFL, LULL, OT) recorded hourly from electrical transformers. We use the standard chronological split: 12 months training, 4 months validation, 4 months test.

Training protocol. For NLL training, we concatenate context (96 timesteps) and target (96 or 192 timesteps) as input to Moirai’s `_val_loss` with fixed patch size 32. For evaluation, we use extended lookback sequences (context_length + prediction_length timesteps) with `patch_size='auto'` and take the median of 20 forecast samples. All data is fed to Moirai in raw scale (Moirai’s internal `PackedStdScaler` handles instance normalization); predictions and targets are then normalized using training-set mean and standard deviation for MSE/MAE computation, matching the standard time-series forecasting evaluation protocol.

Temporal contrastive labels. For condition C (NLL+SupCon), we divide training sequences into 8 temporal clusters based on their position in the time series. Sequences within the same cluster are positive pairs; sequences from different clusters are negatives. This encourages the encoder to maintain temporal-regime awareness.

Horizon selection. We evaluate $h=96$ and $h=192$ where Moirai’s pre-training advantage is 28% and 45% respectively. Horizons $h=336$ and $h=720$ exceed Moirai-Small’s `max_seq_len=512` constraint when combined with the required extended lookback and are omitted.

E ETTh2 Drift Diagnosis and Mitigation (Full Tables)

This section contains the full tables referenced in §4. Table 3 shows the representation drift diagnosis; Table 4 shows the mitigation spectrum.

Table 3: Representation drift diagnosis on ETTh2 forecasting (mean $\pm$ std across 10 seeds). Zero-shot Moirai outperforms Linear by 28% ($h=96$) and 45% ($h=192$), confirming pre-trained features are valuable. Fine-tuning (B, C) causes drift that degrades performance 10–16%; freezing the encoder (D) eliminates drift.

Condition	$h = 96$ (ZS MSE: 0.126)				$h = 192$ (ZS MSE: 0.158)			
	MSE	Forg.%	CKA	Drift	MSE	Forg.%	CKA	Drift
B: NLL-only	.140 $\pm$.014	+11.1	.937 $\pm$.028	3.96 $\pm$.56	.183 $\pm$.020	+15.7	.970 $\pm$.009	4.12 $\pm$.32
C: NLL+SupCon	.138 $\pm$.010	+10.1	.949 $\pm$.014	3.72 $\pm$.43	.178 $\pm$.011	+12.9	.964 $\pm$.015	3.97 $\pm$.34
D: Frozen enc.	.126$\pm$.006	+0.5	1.00$\pm$.000	1.11$\pm$.15	.155$\pm$.002	−1.7	1.00$\pm$.000	1.30$\pm$.18

Table 4: Mitigation spectrum on ETTh2 forecasting. Methods ordered by representation preservation (CKA, $h=96$). LoRA preserves representations (CKA $\approx$ 0.98) while *improving* over zero-shot. Mean $\pm$ std; Forgetting% = (fine-tuned MSE − zero-shot MSE) / zero-shot MSE $\times$ 100.

Method	$h = 96$ (ZS MSE: 0.126)				$h = 192$ (ZS MSE: 0.158)				n
	MSE	Forg.%	CKA	Drift	MSE	Forg.%	CKA	Drift	
B: Full fine-tune	.140 $\pm$.014	+11.1	.937 $\pm$.028	3.96 $\pm$.56	.183 $\pm$.020	+15.7	.970 $\pm$.009	4.12 $\pm$.32	10
F: L2-SP ($\lambda=0.01$)	.150 $\pm$.006	+19.0	.939 $\pm$.023	4.98 $\pm$.34	.179 $\pm$.009	+13.3	.945 $\pm$.007	4.84 $\pm$.22	5
C: NLL+SupCon	.138 $\pm$.010	+10.1	.949 $\pm$.014	3.72 $\pm$.43	.178 $\pm$.011	+12.9	.964 $\pm$.015	3.97 $\pm$.34	10
F: L2-SP ($\lambda=0.1$)	.147 $\pm$.003	+16.7	.965 $\pm$.006	2.94 $\pm$.08	.181 $\pm$.003	+14.6	.980 $\pm$.002	2.84 $\pm$.02	5
G: EWC ($\lambda=1000$)	.149 $\pm$.015	+18.3	.972 $\pm$.004	3.29 $\pm$.79	.188 $\pm$.004	+19.0	.978 $\pm$.006	4.46 $\pm$.38	5
E: LoRA ($r=8$) [†]	.116$\pm$.001	−7.9	.979 $\pm$.006	0.00$\pm$.00	.157$\pm$.002	−0.6	.986 $\pm$.005	0.00$\pm$.00	5
D: Frozen encoder	.126 $\pm$.006	+0.5	1.000$\pm$.000	1.11 $\pm$.15	.155 $\pm$.002	−1.7	1.000$\pm$.000	1.30 $\pm$.18	10

[†] LoRA ($\alpha=16$) adds low-rank updates that do not modify base encoder parameters; ℓ_2 drift is exactly zero by construction.

F LoRA Rank and Hyperparameter Sensitivity

Table 5 shows LoRA on **Moirai-Small** is robust to rank ($r \in \{4, 8, 16, 32\}$): all ranks achieve CKA $\geq$ 0.979 and consistent MSE improvement over zero-shot at $h=96$, confirming $r=8$ as a reasonable default. On **Moirai-Large** (seed 42, $h=96$, $n=500$), rank escalation alone does *not* rescue the failure mode: forgetting stays +19 to +30% across $r \in \{8, 16, 32, 64\}$. Table 6 extends the sweep to α , target-module scope, and LR (all at $r=8$, seed 42): varying $\alpha \in \{8, 16, 32\}$ and module scope provides no rescue, but reducing LR to 1×10^{-5} does (forg.= $-8.5 \pm 0.7\%$, CKA=0.989 $\pm$ 0.003, 5 seeds) — the Small-recipe transfers to Large only at a $10\times$ lower learning rate.

G CKA Calibration: Random-Initialisation Baseline

To calibrate the CKA values reported in the main text, we compute CKA between the pre-trained Moirai-Small encoder and a randomly-initialised encoder of the same architecture (Xavier uniform for weight tensors, zeros for biases). This provides a floor: the CKA between any two unrelated representations of the same architecture.

Using 50 benign probe samples and 5 random seeds (42, 123, 456, 789, 999), the random-initialisation CKA baseline is reported in Table 7.

The random-init CKA is $\leq 10^{-4}$ across 5 seeds (reported at 4 decimals as 0.0000 ± 0.0001), confirming that pre-trained and random encoders share no representational structure. This is consistent with the theoretical expectation: CKA between an orthogonally-initialised encoder and any fixed reference encoder has expectation asymptotically zero in the feature dimension. Since IoT fine-tuned CKA (0.31–0.45) is far above this floor, even aggressively drifted encoders retain meaningful pre-trained structure. ETT fine-tuned CKA (0.81–0.94) is near-ceiling, confirming substantial

Table 5: LoRA rank sensitivity on ETTh2 (condition E, 500 training samples). Moirai-Small: all ranks achieve near-identical MSE and $\text{CKA} \geq 0.979$. Moirai-Large: all ranks forget +19 to +30%, with no rank recovering Moirai-Small’s negative forgetting. Mean $\pm$ std across 3 seeds (5 for $r=8$ Small); Moirai-Large single-seed ($s=42$) for $r \in \{16, 32, 64\}$, 3 seeds for $r=8$.

Backbone	Rank	h	MSE	Forg.%	CKA	Seeds
Small	4	96	.116 $\pm$.001	−9.5 $\pm$ 0.9	.992 $\pm$.001	3
Small	4	192	.146 $\pm$.003	−8.8 $\pm$ 1.9	.995 $\pm$.002	2
Small	8	96	.116 $\pm$.001	−9.1 $\pm$ 1.1	.979 $\pm$.006	5
Small	8	192	.157 $\pm$.002	−1.8 $\pm$ 3.1	.986 $\pm$.005	5
Small	16	96	.117 $\pm$.001	−10.2 $\pm$ 1.2	.987 $\pm$.002	3
Small	16	192	.155 $\pm$.002	−1.7 $\pm$ 1.2	.988 $\pm$.001	3
Small	32	96	.119 $\pm$.001	−8.3 $\pm$ 0.9	.986 $\pm$.001	3
Small	32	192	.159 $\pm$.003	+1.1 $\pm$ 1.4	.989 $\pm$.002	3
Large	8	96	.159 $\pm$.004	+25.7 $\pm$ 3.2	.973 $\pm$.012	3
Large	16	96	.154	+22.5	.955	1
Large	32	96	.150	+19.0	.986	1
Large	64	96	.165	+29.9	.984	1

Table 6: LoRA-Large HP ablation (ETTh2, $r=8$, $n=500$, $h=96$). Baseline (default LR=1e-4, $\alpha=16$, modules = q,v,out) forg.= +22.5%, seed 42. Only LR=1e-5 rescues LoRA-Large (5 seeds); all other axes fail.

Axis	Setting	Val MSE	Forg.%	CKA	Seeds
LR	1e-4 (default)	.157	+22.5	.955	1
	5e-5	.146	+15.3	.982	1
	1e-5	.115$\pm$.001	−8.5$\pm$0.7	.989$\pm$.003	5
α	8	.155	+21.9	.975	1
	16 (default)	.157	+22.5	.955	1
	32	.158	+24.3	.984	1
Modules	q,v,out (default)	.157	+22.5	.955	1
	q,k,v	.153	+20.4	.987	1
	q,k,v,out	.151	+19.0	.988	1

preservation. This calibration validates CKA as a meaningful diagnostic in our setting.

H Linear Probing: Functional Representation Diagnostic

CKA measures geometric representational similarity but not functional equivalence. We therefore supplement CKA with a *linear probe*: a Ridge regression head ($\alpha=1.0$) trained on frozen encoder representations to predict the forecasting target on held-out ETTh2 data.

Protocol. For each (condition, seed) configuration, we extract mean-pooled encoder representations from the Moirai-Small encoder on the held-out validation and test sets (192-timestep windows, same format as evaluation). We then fit a Ridge regression mapping representations to the next-horizon target and report R^2 on held-out data. The probe training set uses $n_{\text{probe}}=300$ held-out examples; the remainder forms the test split used for MSE evaluation. ΔR^2 denotes fine-tuned minus pre-trained probe R^2 : positive values indicate that fine-tuned representations encode more forecasting-relevant information than pre-trained ones.

Interpretation. Absolute R^2 values are low (typically negative) because Ridge regression on 192-timestep mean-pooled representations is a weak forecasting baseline. The *relative* change (ΔR^2) is the meaningful signal: it isolates whether fine-tuning added or removed task-relevant information

Table 7: CKA calibration. Random-init CKA (0.000 ± 0.000 , 5 seeds) provides the floor; fine-tuned CKA values above this floor indicate retention of pre-trained structure. IoT fine-tuning (CKA 0.31–0.45) retains meaningful structure despite aggressive drift. ETT fine-tuning (CKA 0.81–0.94) preserves most pre-trained structure.

Comparison	CKA
Pre-trained vs. Random-init (baseline, 5 seeds)	0.0000 ± 0.0001
Pre-trained vs. ETT fine-tuned (B, $h=96$)	0.937–0.964
Pre-trained vs. IoT fine-tuned (C, 200/5, 10 seeds)	0.420 ± 0.026
Pre-trained vs. Frozen encoder (D)	1.000

from the encoder’s representational basis.

Key result. The sample-size sweep (Table 1) shows ΔR^2 monotonically increasing with training data (Ridge: $+0.12$ at 500 samples $\rightarrow +0.67 \pm 0.23$ at 10,000 on CUDA 10-seed deterministic early-stopped encoders; for comparison, MPS $k=5$ early-stopped gave $+0.56 \pm 0.14$ and final-epoch $k=5$ gave $+0.78 \pm 0.14$), while CKA decreases monotonically. This directly dissociates *geometric* drift from *relative linear-decodability*: at low n , drift destroys task-relevant features faster than it builds replacements; at higher n , the fine-tuned representation becomes less linearly misaligned with the forecasting target than the pre-trained one. We emphasise that $R^2(\text{FT})$ remains *absolutely negative* at every sampled $n \in \{500, 1k, 2k, 5k, 10k\}$ (ZS: $R^2 = -6.90$; CUDA $k=10$ early-stopped: -6.79 at $n=500 \rightarrow -6.24 \pm 0.24$ at $n=10k$). $\Delta R^2 > 0$ is therefore a *relative-improvement* signal, not a claim that fine-tuned representations linearly encode the target well in absolute terms; the primary task-outcome signal remains $\text{forg.} < 0$ at $n \geq 2k$. **Cross-dataset ΔR^2 comparability.** Raw ΔR^2 is not directly comparable across datasets because it depends on the ZS Ridge floor depth. On ETTm2 ($n=10k$, CUDA $k=10$), the ZS Ridge floor is $R^2(\text{ZS}) = -24.52 - 3.5 \times$ deeper than ETTTh2’s -6.90 — which proportionally inflates the raw ΔR^2 ($+5.43$ on ETTm2 vs. $+0.67$ on ETTTh2). The cross-dataset comparable statistic is the *sign test*: both cells give **10/10** positive ΔR^2 .

Horizon-sweep probe: head-specificity of the ΔR^2 gain. Table 8 reports Ridge ΔR^2 across three probe heads on four early-stopped encoders at $n=10k$ (the same encoders as the main-text MLP sweep). The gain is concentrated on the *trained* 96-step head: $\Delta R^2 = +0.116 \pm 0.153$ ($k=4$). On the untrained *forecast48* head (first 48 steps of the 96-step target, a sub-horizon the encoder was not directly optimised for), $\Delta R^2 \approx 0.000 \pm 0.163$ — the gain vanishes at the horizon boundary. On the untrained *delta1* head, a GBM probe gives $R^2(\text{PT}) = +0.059 > 0$, establishing a positive-floor reference, with $\Delta R^2 = -0.056 \pm 0.030$ (decodability *falls* on the untrained readout). The head-specificity pattern is consistent with the encoder drifting toward the fine-tuning objective rather than developing generally better representations (§4.4).

Table 8: Ridge ΔR^2 by probe head on early-stopped encoders ($n=10k$, seeds {303, 777, 888, 999}). “Forecast96” is the trained head; “Forecast48” and “Delta1” are untrained. The gain is head-specific to the trained objective.

Head	Trained?	Ridge ΔR^2 per seed			
		s303	s777	s888	s999
Forecast96 (96-step ahead)	✓	+0.092	−0.087	+0.266	+0.192
Forecast48 (sub-horizon)	×	+0.061	−0.111	−0.156	+0.199
Delta1 (next-step delta)	×	+1.248	+0.757	+1.191	+0.980
Mean $\pm$ std (forecast96)			$+0.116 \pm 0.153$		
Mean $\pm$ std (forecast48)			$−0.001 \pm 0.163$		
Mean $\pm$ std (delta1)			$+1.044 \pm 0.223$		

Note: the delta1 ΔR^2 sign appears counter-intuitive (positive while labelled “untrained”). The large positive delta1 ΔR^2 reflects that the ZS delta1 Ridge baseline is very negative ($R^2(\text{ZS}) \approx -2.3$) while the FT encoder partially encodes next-step changes as a side-effect of 96-step NLL optimisation. The diagnostic signal is the GBM result (§4.4): GBM gives $R^2(\text{PT}) = +0.059 > 0$ and $\Delta R^2 = -0.056$,

yielding $R^2(\text{FT})=+0.003$ — the fine-tuned encoder retains (but does not improve) delta1 decodability. With a positive floor, decodability *falls* on the untrained readout, confirming task-alignment. Note: $R^2(\text{PT})$ varies slightly across GBM probe depths (depth 4: +0.062, depth 6: +0.059, depth 8: +0.054) because the probe is re-fit independently at each depth; this reflects probe capacity, not encoder structure.

ETTh2 delta1 GBM reprobe. We ran the delta1/GBM probe on the five confirmed ETTh2 $n=10\text{k}$ encoders (seeds 42/101/123/202/303) to check for a positive-floor + $\Delta R^2 > 0$ cell on a second dataset. Result: $R^2(\text{ZS})=-0.030$ (negative floor) and per-seed $\Delta R^2=\{+0.025, -0.060, +0.026, +0.030, +0.016\}$ (4/5 positive, mean $+0.007 \pm 0.031$). *Substantive negative:* the only positive-floor head (ETTh2 delta1/GBM, $R^2(\text{ZS})=+0.059$) gives $\Delta R^2=-0.056$ — no positive-floor + $\Delta R^2 > 0$ cell was found. The $\Delta R^2 > 0$ signal on the trained 96-step head operates in a deeply-negative- R^2 regime; see Appendix K for probe noise-floor analysis.

Per-seed breakdown at $n=10\text{k}$ (CUDA 10-seed). Ridge ΔR^2 (all positive; mean $+0.668 \pm 0.226$): 42: +0.205; 101: +0.736; 123: +0.937; 202: +0.604; 303: +0.793; 456: +0.823; 777: +0.344; 789: +0.873; 888: +0.561; 999: +0.804. Absolute $R^2(\text{FT})$: all in $[-6.70, -5.97]$ ($R^2_{\text{ZS}} = -6.904$). MLP $k=5$ (all positive; mean $+0.701 \pm 0.324$): 42: +0.979; 101: +0.762; 123: +0.948; 202: +0.771; 303: +1.007; 456: +0.266; 777: +0.076; 789: +0.847; 888: +0.449; 999: +0.901.

ETTh2 $n=10\text{k}$ absolute R^2 per seed. The ETTh2 ZS Ridge floor ($R^2(\text{ZS}) = -24.52$, same pre-trained encoder across all seeds) is $3.5\times$ deeper than ETTh2's -6.90 , explaining the raw ΔR^2 magnitude difference. Per-seed absolute values for the five available CUDA seeds:

Seed	$R^2(\text{ZS})$	$R^2(\text{FT})$	ΔR^2	Forg.%
42	-24.52	-14.13	+10.39	-26.8
101	-24.52	-15.61	+8.91	-27.4
123	-24.52	-19.87	+4.64	-31.1
202	-24.52	-20.83	+3.69	-28.8
303	-24.52	-20.41	+4.11	-29.5
Mean (5 seeds)	-24.52	-18.17 $\pm$ 3.08	+6.35 $\pm$ 3.08	-28.7 $\pm$ 1.6

All five seeds show $R^2(\text{FT}) > R^2(\text{ZS})$ (i.e., fine-tuned representations are *less* linearly misaligned with the ETTh2 forecasting target than pre-trained ones), and all show $\Delta R^2 > 0$. The cross-dataset comparable statistic is the sign test: both ETTh2 (**10/10**) and ETTh2 (**5/5** seeds: 42, 101, 123, 202, 303) give unanimous $\Delta R^2 > 0$.

MLP depth sweep on early-stopped encoders. A cosine-LR re-run at $n=10\text{k}$ uses seeds {303, 777, 888, 999} at epochs=10 (seed 101 MPS-NaN'd on epoch 3 reproducibly at this schedule and is excluded). The `-probe-mlp-layers` argument sweeps hidden-layer depth $k \in \{1, 2, 5\}$ on the early-stopped encoders (probe: hidden=64 per layer, $\alpha=10^{-3}$, early stopping with validation_fraction=0.1, max_iter=500):

Seed	Ridge ΔR^2	MLP $k=1$	MLP $k=2$	MLP $k=5$
303	+0.092	+1.306	+1.418	+0.540
777	-0.087	+1.136	+1.554	+0.759
888	+0.266	+2.004	+2.361	+1.034
999	+0.192	+1.816	+1.949	+0.908
Mean $\pm$ std	+0.116 $\pm$ 0.132	+1.566 $\pm$ 0.356	+1.821 $\pm$ 0.368	+0.810 $\pm$ 0.184

The cosine-LR Ridge ΔR^2 is smaller than the epochs=20 run ($+0.56 \pm 0.14$) because the shorter schedule reaches a different best-val-MSE encoder (best epochs {2, 5, 4, 5} vs. {2, 4, 8, 4, 3}), and Ridge is most sensitive to the specific linear basis. The non-monotonic $k=1 \rightarrow k=2 \rightarrow k=5$ trend

Table 9: Per-seed results at $n=5,000$ (ETTh2, $h=96$, condition B, 20 epochs). Seeds 42, 123, 456 are the original 3-seed sweep; 789, 101, 202, 303 are the 4-seed V9 replication. Forgetting is bimodal; ΔR^2 is uniformly positive.

Seed	MSE(FT)	Forg.%	CKA	ΔR^2	R^2 (FT)
42	.132	+4.8	.531	+.67	−6.24
101	.120	−4.9	.571	+.75	−6.16
123	.124	−1.4	.544	+.81	−6.10
202	.128	+1.4	.461	+.15	−6.76
303	.150	+19.0	.525	+.52	−6.39
456	.153	+21.4	.711	+.20	−6.70
789	.134	+5.9	.480	+.40	−6.51
Mean±std	.134±.012	+6.6±10.0	.546±.082	+.50±.26	−6.41±.26

reflects the deeper probe fitting the pre-trained encoder better (per-seed mean $R^2(\text{PT})_{k=5} = -6.93$ vs. $R^2(\text{PT})_{k=1} = -8.48$), shrinking the Δ from above, not degraded fine-tuned encoding. All four seeds are individually positive for $k=5$ MLP (+0.54, +0.76, +1.03, +0.91), preserving the cost-benefit gap. This run’s CKA is 0.592 ± 0.142 (mean best epoch 4.0 ± 1.22 , higher than the earlier run’s 4.2 ± 2.3 because of the earlier LR annealing); forgetting is $-7.13 \pm 1.34\%$, tighter than the epochs=20 run’s $-11.7 \pm 5.3\%$ and fully consistent with the $\text{forg.} < 0$ finding.

Crossover regime: per-seed breakdown at $n=5,000$. Table 9 reports all 7 seeds individually at $n=5,000$. Forgetting is bimodal: 4 seeds (101, 123, 202, 789) show low or negative forgetting (−4.9% to +5.9%, mean +0.7%) and 3 seeds (42, 303, 456) show high forgetting (+4.8% to +21.4%, mean +15.1%). All 7 seeds show positive ΔR^2 (+0.15 to +0.81, mean +0.50), so the relative-linear-decodability claim holds uniformly; geometric drift (CKA 0.46–0.71, mean 0.55) varies less than forgetting. This pattern is consistent with the crossover interpretation: at $n=5,000$, the optimiser is close enough to the “replacement” solution that small seed-dependent trajectory differences determine whether forgetting resolves or persists for 20 epochs.

H.1 $n=5,000$ per-epoch trajectory analysis

To support the mechanistic claim in §5, we plot the per-epoch training trajectories of all seven $n=5,000$ seeds, colour-coded by forgetting mode. Figure 2 shows val MSE and weight drift across training. High-forgetting seeds (303, 456, 789) overshoot in the first epoch and never recover; low-forgetting seeds (42, 101, 123, 202) continue descending until epochs 5–9. The epoch at which val MSE reaches its minimum separates the modes (0.7 ± 0.5 vs. 6.5 ± 2.1); mid-training weight drift (epochs 5–10) does not. Code: `scripts/analyse_n5k_trajectories.py`.

I ETTh1 Extended Analysis

Spectral entropy pilot (post-hoc, not pre-specified). We computed spectral entropy (Welch PSD, normalised Shannon entropy) and the Hurst exponent (rescaled-range) on the OT (oil temperature) target column of each ETT dataset: ETTh1 (SE=3.61 bits, $H=0.954$), ETTh2 (SE=3.46 bits, $H=0.946$), ETTm2 (SE=2.25 bits, $H=0.941$). ETTh1 has the highest spectral entropy (most broadband power, most trend-dominated) and is the only dataset where $\Delta R^2 > 0$ does not hold at $n=10k$; ETTm2 has the lowest (most concentrated, most cyclical) and shows the strongest $\Delta R^2 > 0$. ETTh1 and ETTm2 have similar deep ZS Ridge floors ($R^2(\text{ZS}) \approx -25$) yet opposite ΔR^2 signs, ruling out floor depth as a predictor on its own. The spectral-structure pilot is consistent with a trend-vs.-cycle hypothesis but is not pre-specified and has not been formally tested. A pre-registered test on held-out datasets is the appropriate next step. These numbers were computed at a prior

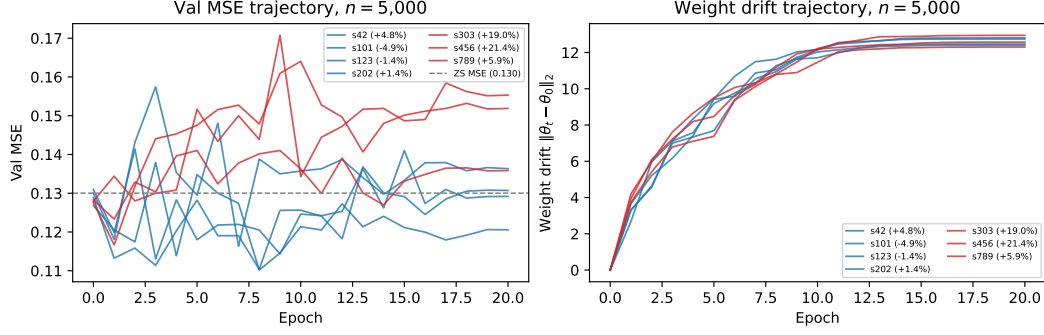


Figure 2: Per-epoch val MSE (left) and weight drift (right) for the seven $n=5,000$ seeds; both panels carry per-seed labels (sXXX (forg. %)) so each trajectory is individually identifiable. Blue: low-forgetting mode (forg. $\leq 5\%$); red: high-forgetting mode (forg. $> 5\%$). Grey dashed line: zero-shot MSE. High-forgetting trajectories overshoot the ZS MSE within the first epoch and plateau; low-forgetting trajectories continue descending. Mid-training weight drift (epochs 5–10) does not separate the modes.

reviewer’s request; the methodology and code are available in the project repository.

J IoT Negative-Control Details

The IoT negative-control evaluation uses CICIOT2023 [13] with 12 network-flow features and 128-timestep windows; the anomaly score is the fraction of timesteps falling outside the 95% CI of Moirai’s forecast over a 32-timestep scored horizon. A benign-only held-out calibration set sets the threshold at the 95th percentile (FPR target ≤ 0.05). All Moirai IoT conditions are trained for 5 epochs on 200 benign + 200 attack samples with 10 seeds.

CNN-NLL from-scratch control. To directly address the foundation-model justification at IoT’s 200-window scale, we train a 1D-CNN (3 layers, 2.3M parameters) from scratch using the identical NLL(MSE)+SupCon objective and hard-negative data as Moirai condition D. At 10 seeds, CNN-NLL ($\text{AUC}=0.598\pm0.041$) matches fine-tuned Moirai B/D ($\text{AUC}=0.556\pm0.021$) and is close to the best Moirai condition C ($\text{AUC}=0.629\pm0.022$). A CNN-MSE variant ($\text{AUC}=0.580\pm0.023$) rules out the loss-function confound. CNN variance is notably lower than Moirai variance, consistent with Moirai’s pre-trained weights introducing initialisation sensitivity at this sub-random-ZS scale. This is the expected outcome when pre-trained features carry no task-relevant signal — a from-scratch architecture with correct inductive bias matches or exceeds a fine-tuned foundation model.

K Probe Noise-Floor Analysis

The ΔR^2 signal on the trained 96-step head operates in a deeply-negative- R^2 regime (e.g., ZS -6.90 , FT -6.24 at $\alpha=1$). We test whether the $\Delta R^2>0$ direction is an artefact of probe-design choices by running Ridge on two fixed encoders (ZS and condition-B seed 42, ETTh2, $n=10k$) across: (1) $\alpha \in \{0.01, 0.1, 1.0, 10.0, 100.0\}$, (2) probe seeds $\{0, 1, 2\}$, (3) mean vs. last-token pooling. No retraining required (saved `best_encoder.pt`).

Results (Table 10).

Finding: $\Delta R^2>0$ holds for all 10 probe-design settings tested; the direction is not an artefact of α , pooling, or probe seed. Magnitude decreases with stronger regularisation (larger α shrinks both R^2 values toward zero), but the sign is invariant. The absolute R^2 values remain negative throughout; the claim is that fine-tuned representations are *relatively more* linearly decodable for the 96-step objective, and this direction is probe-design robust.

Table 10: ΔR^2 under probe-design perturbations (ETTh2 96-step, ZS vs. FT seed 42). All $\Delta R^2 > 0$.

Setting	$R^2(\text{ZS})$	$R^2(\text{FT})$	ΔR^2
$\alpha=0.01$	-13.51	-11.54	+1.96
$\alpha=0.1$	-8.78	-7.94	+0.84
$\alpha=1.0$ (default)	-6.90	-6.70	+0.20
$\alpha=10$	-5.81	-5.72	+0.08
$\alpha=100$	-5.32	-5.28	+0.04
Probe seeds 0/1/2 ($\alpha=1$, mean pool)	same as $\alpha=1.0$		+0.20
Last-token pool ($\alpha=1$)	-9.70	-8.49	+1.20

L Random-Initialisation Negative Control

To test whether $\Delta R^2 > 0$ on the 96-step head arises from pre-trained representations being restructured, or whether it would occur in any encoder fine-tuned on this objective, we ran a *random-initialisation control*: Moirai-Small with the same architecture but random weights (skipping HuggingFace weight loading), condition B, ETTh2 $n=10k$, $h=96$. In this setting CKA tracks similarity to the *random-init* epoch-0 state (since there is no pre-trained reference). We ran 5 seeds (42/101/123/303/456) matching a subset of the condition B seed set.

Results (seeds 42/101/123/303/456):

Seed	CKA	$R^2(\text{PT})$	$R^2(\text{FT})$	ΔR^2	ZS MSE $\rightarrow$ FT MSE
42	0.357	-6.554	-6.692	-0.138	0.140 $\rightarrow$ 0.378
101	0.505	-6.780	-6.631	+0.149	0.164 $\rightarrow$ 0.398
123	0.540	-6.749	-6.944	-0.195	0.153 $\rightarrow$ 0.354
303	0.318	-6.834	-7.239	-0.405	0.151 $\rightarrow$ 0.359
456	0.273	-6.665	-7.070	-0.405	0.160 $\rightarrow$ 0.360
Mean $\pm$ std	0.399 $\pm$ 0.105			-0.199 $\pm$ 0.205	

Across 5 seeds: $\Delta R^2 = -0.199 \pm 0.205$ (4/5 negative). The one positive seed (+0.149) is substantially smaller in magnitude than condition B’s per-seed range (+0.38 to +0.75 for early-stopped seeds). None of the 5 random-init seeds approaches the pre-trained condition B mean (+0.668 $\pm$ 0.226, 10/10 positive).

Compare to condition B (pre-trained): $\Delta R^2 = +0.363$ at seed 42 (and +0.668 $\pm$ 0.226 across $k=10$ CUDA seeds). The random-init mean $\Delta R^2 = -0.199$ is substantially negative versus the pre-trained encoder’s consistently positive signal. This confirms that the robust $\Delta R^2 > 0$ pattern is not a property of any encoder fine-tuned on the 96-step NLL objective; the pre-trained structure is required for the consistent positive directional shift. The decodability shift toward the objective reflects restructuring of pre-trained representations, not incidental training noise.

M Layer-Wise Unfreeze Localization

Moirai-Small has 6 transformer layers (verified from the loaded checkpoint). We added an `-unfreeze-top-n-layers` flag to `finetune_forecasting.py` and ran ETTh2 $n=10k$, $h=96$, 3 seeds (42/101/123) at $N=3$ alongside the existing $N=0$ (frozen encoder, 5-seed condition D) and $N=6$ (full unfreeze, condition B; $N=6$ unfreezes all 6 layers).

$N=1$ instability. At $N=1$, training diverged (NaN activations at epoch 11, batch 148) with default $\text{LR} = 10^{-4}$, despite gradient clipping at norm 1. The 11 epochs that completed show CKA stabilising at ≈ 0.72 ; we do not extract probe results from a crashed run. This instability is qualitatively consistent

with the LoRA-Large behaviour documented in §4.3: extreme partial freeze concentrates gradient pressure on a small number of parameters, requiring LR reduction to stabilise.

$N=3$, $N=4$, $N=5$ **results.** Table 11 shows results for all six values of N : $N=0$ (5 seeds), $N=3$ and $N=6$ (3 matching seeds), and $N=4$ and $N=5$ (1 seed 42 each).

Table 11: Layer-wise unfreeze profile (ETTh2 $n=10k$, $h=96$; mean $\pm$ std over seeds). N = number of top transformer layers unfrozen (Moirai-Small has 6 layers). $N=0$: 5 seeds; $N=3/N=6$: seeds 42/101/123; $N=4/N=5$: seed 42 only. Forgetting is final-epoch (no early stopping); $N=5$ overshoot reflects late-epoch divergence.

N	Layers	CKA	Ridge ΔR^2	Forgetting%
0 (frozen)	none	0.999 ± 0.000	-0.048 ± 0.008	$-6.6\pm 1.5\%$
1 (top-1)	1/6	≈ 0.72 (ep. 11)	<i>diverged (NaN)</i>	
3 (top-3)	3/6	0.698 ± 0.066	$+0.479\pm 0.115$ (3/3)	$+1.3\pm 2.1\%$ [†] Final-epoch overshoot;
4 (top-4)	4/6	0.610 (1 sd)	$+0.363$ (1/1)	$+4.1\%$
5 (top-5)	5/6	0.644 (1 sd)	$+0.767$ (1/1)	$+31.8\%$ [†]
6 (full, B)	6/6	0.484 ± 0.061	$+0.626\pm 0.378$ (3/3)	$-7.7\pm 8.4\%$

best-epoch ($h=96$, epoch 7) MSE=0.132 $\approx$ ZS.

Finding: ΔR^2 and forgetting dissociate across unfreeze depth. $\Delta R^2 > 0$ appears already at $N=3$ (half the encoder, 3/3 seeds, CKA ≈ 0.70), with substantially less geometric drift than full unfreeze (CKA ≈ 0.48). $N=4$ and $N=5$ (1 seed each) both give $\Delta R^2 > 0$ (1/1), confirming the decodability signal extends through the full $N=3$ –6 range. The ΔR^2 values at $N=4$ (+0.363) and $N=5$ (+0.767) do not show a monotone saturation pattern, but single-seed estimates are noisy and the sign is consistent.

The key dissociation is in *forgetting*: $N=3$ gives forg. $+1.3\pm 2.1\%$ (task slightly worse than ZS), $N=4$ gives $+4.1\%$ (1 seed), and $N=6$ (full unfreeze, condition B) gives $-7.7\pm 8.4\%$ (task substantially better). The $N=5$ final-epoch forgetting of $+31.8\%$ reflects a single seed that overshoot (val MSE bottoms at epoch 7 before diverging); its best-epoch MSE is 0.132, essentially unchanged from ZS—qualitatively consistent with the $N=5k$ trajectory analysis (§??). ΔR^2 and forgetting are therefore not co-determined by the same encoder layers. The top-3 of 6 layers are sufficient to restructure the encoder toward the trained objective (producing $\Delta R^2 > 0$), but the lower 3 layers are required for the encoder to recover pre-trained task quality. The additional ΔR^2 gain from $N=3$ to $N=6$ ($+0.48 \rightarrow +0.63$) is modest and has much higher variance ($0.115 \rightarrow 0.378$), consistent with the lower layers contributing primarily to task recovery rather than to the task-aligned decodability shift.

This finding sharpens the central claim: the frozen-encoder control (Appendix K) rules out data-distribution confound; the $N=3$ result rules out a whole-encoder-restructuring account. The decodability shift toward the objective is localised to the top transformer layers, and $\Delta R^2 > 0$ is a *necessary but not sufficient* predictor of task improvement across unfreeze levels. Both $N=3$ and $N=6$ operate in the same deeply-negative- R^2 regime; the probe-design robustness of Appendix K applies to both.

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